Lab Experiment 9

Electronics and computer engineering software

Learning Outcomes

By the end of these activities, students will be able to:

1. Apply **Ohm's Law** to calculate the required resistor value for an LED given supply voltage, forward voltage, and current.

2. **Differentiate between series and parallel circuits** and explain why parallel is preferred for powering multiple LEDs.

3. **Design and simulate a working LED circuit** using TinkerCAD, ensuring correct polarity, current-limiting resistors, and safe operation.

4. **Expand a circuit to include multiple** while managing shared and independent power paths.

5. **Integrate a switch** that controls only a sub-section of the circuit (LEDs) without affecting the sensor's operation.

Lab Exercise

Objective:

To introduce students to the fundamentals of circuit design and analysis using both theoretical calculations and hands-on simulation. Students will apply Ohm's Law, design and test circuits with proper current-limiting resistors, and expand the designs to include multiple loads and extra component implementation.

Instructions

Complete the following activities referring to the lab manual, lesson slides, chapter 6, TinkerCAD, and other appropriate resources.

Lab Activities:

Part A.

You are given the following single LED circuit:

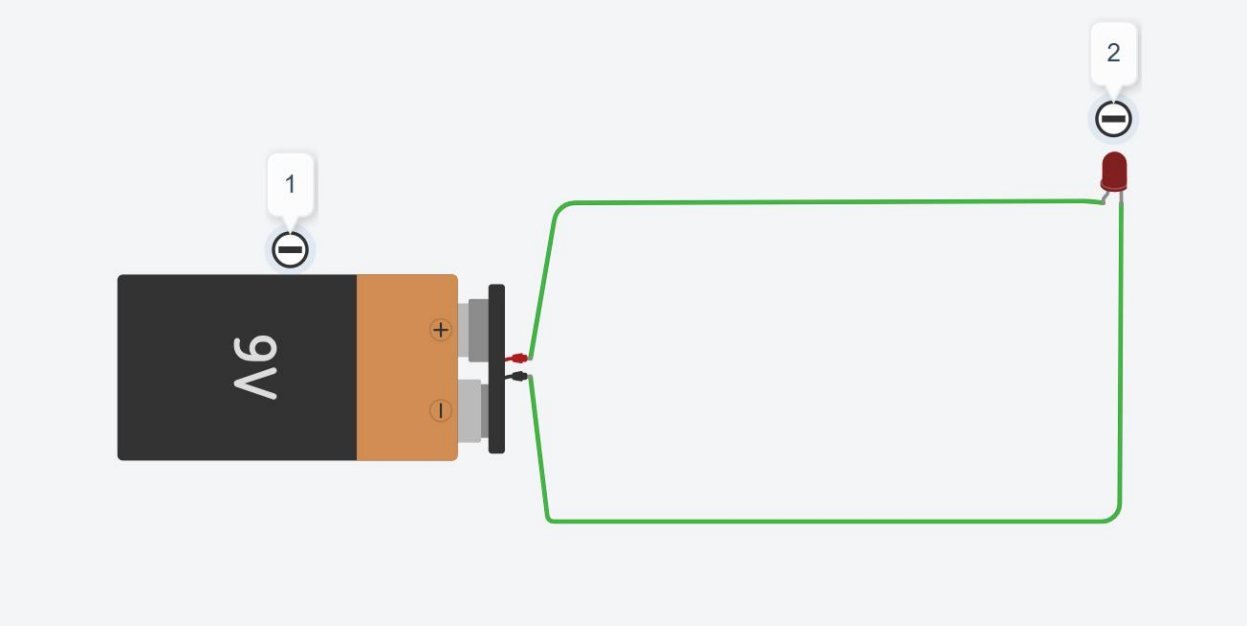


Figure 1 Basic circuit

The specifications for each component are:

$$V_1 = 9V$$

$$V_2 = 2V$$

Questions:

1. If the current flowing through the circuit is equal to $I = 20mA$, calculate the **resistance R** needed for the LED to safely operate at a voltage of **2V**. Once you are done calculating the resistance, create the same circuit on your TinkerCAD account and add the resistance, please provide a screenshot on your lab report (You can use the windows snipping tool).

2. **Suppose** we want to add **2 more LEDs** to this circuit, each with a voltage drop of 2V and a total current I of 20mA. Supplied voltage is the same (9V)

$$V_1 = 2V$$

$$V_2 = 2V$$

$$V_3 = 2V$$

$$I = 20mA$$

You have two options, either connect them in **parallel** or in **series**.
For this question, answer the following in your lab report:

- **Explain** what happens to the voltage drop and current for each case.
- **Calculate** the total voltage drop and current for each case.
- **Discuss** the advantages and disadvantages of each case, you would have to choose one of the cases to solve the remaining questions on this lab.

3. **Modify** your current circuit on TinkerCAD to the design choice you made in question #2, your circuit must include the 3 LEDs, the voltage supply (battery), and the appropriate resistances in either a series or parallel arrangement.
   Hint: You would only need 1 resistance if you chose series for question #2, and 3 resistances (1 for each branch) if you chose parallel.

4. For unknown reasons, the voltage supply in your circuit becomes unstable; to improve the safety and reliability of your circuit, you need to expand it by adding the following components to it:

   - Switch
   - Capacitor

   - **Add** a switch to your current design and provide a screenshot on your lab report.
   - There are different types of capacitors to use, research online about **ceramic** and **electrolytic** capacitors. After that, choose the best one to use in your circuit, taking into account the unstable voltage supply situation. You don't have to add any capacitors to your sketch or perform any capacitance calculation for this task, just provide an explanation.

5. Recreate your full circuit in TinkerCAD, but this time with a breadboard. Not just direct connections, just need to provide a screenshot of your sketch for this part.

Deliverables

You must submit a lab report following the lab template. The lab report must include the work done for each question as well as the usual reflection at the end, plus applicable references.

Grading Rubric

- **Objectives:** Identifies objectives and provides the necessary background or context (0-3 points)
- **Results/Analysis:** Presents results clearly. Analyzes findings correctly, with proper calculations and/or comparisons as needed. (0-6 points)
- **Discussion & Reflection:** Explains significance of results, discusses errors and connects outcomes to engineering/technology concepts. (0-5 points)
- **Professionalism & Format:** Report is well-organized, follows the provided template, uses clear language, proper grammar and formatting. Includes citation and references (if outside sources were used) (0-6 points)
- **Total:** 20 points